Further Lean proof simplification of Wang tile constructions

DiophantineConsistency fourth review

26 October 2026 Pinned source `e652d21c413de59e4821f6e8e9b6a3a077bf5bfe`

Result. One further candidate survives the closing overlap review: a shared halted-color witness for pass and finish tiles. Its complete file would shrink from **164 to 155 physical lines**, a net reduction of **9 lines**, including the new private helper and documentation. The candidate remains uncompiled.

ID	Remaining obligation	Net lines removed	Added API
B2	Derive both halted closing constructors from one color witness	9	One private theorem
	Total over one complete file	**9**	**No definitions or imports**

Late overlap correction. A packet received at 17:54 UTC, *Four Further Lean Proof Refinements*, already proposes period closure for the same helical transport theorem (its C) and the same exact canonical-band quotient helper (its D). Accordingly, H1 and H2 from the initial shortlist are **excluded from new contributions**. Their variants are retained solely in comparison artifacts. A different periodicity API or proof spelling is not a new idea under the requested baseline.

Recommended order. Try B2 as a local private-helper refactor retaining both public endpoints. B1 is additionally excluded because packet w already proposes cyclic shared-edge grids covering the temporal matching obligation. No credit is claimed for excluded comparisons.

Novelty baseline. The original audit grants all 21 placed packets a–u at the source pin, the historical incoming record, and this series' six prior-round ideas. The late packet, which records coverage through v and closing pin `f42d3f94f78eb63639316a33e25f2f69d12c651e`, is an additional exclusion check, not a claimed fresh scan of the entire later repository.

Separate observed progress. The 33 newly incorporated commits already contain four refactor groups saving 1,771 net Lean lines, with helper and lower-dependency costs included. New construction and check modules make the active Lean subtree grow by 1,073 lines overall. Neither the earlier savings nor that new functionality is credited to this report.

Validation boundary. Every new Lean candidate remains **uncompiled**. Source identities, candidate replay, signatures, exact definitions, and bounded models were checked. These are not elaboration, kernel, axiom-footprint, machine-halting, or performance results. No Lean installation or upstream edit was performed. The remainder gives exact proofs to test, preserved contracts, and a reproducible evidence trail.

1 Recent source changes

The reviewed head is `e652d21c413d`, recorded at 17:28:55 UTC on 26 October 2026. The comparison pin is `22e020937caa`. GitHub reports 33 commits ahead and none behind. Subtracting the previous pin's ancestors from the returned commit DAG independently yields the same 33. Date-order truncation would miss two newly incorporated proposal commits whose timestamps precede the earlier pin.

Those commits comprise 12 merges, eight proposal intakes, four adopted refactors, five new-construction commits, and four verification or review records. An intake or archived check is not maintained-source adoption. Upstream records of Lean checks are historical evidence; this review did not rerun them.

1.1 Adoptions already present and excluded from new credit

Commit	Observed refactor	Net lines removed
`79bc166a51`	Checked-label generated Steps	1,646
`32b108f214`	Shared nine-coordinate transport	26
`2effe6f291`	Radix, convolution, selector and carry cleanup	22
`8ca6455829`	Ordinary and doubled Pell ratio estimate	77
	Net including helper costs	**1,771**

The nine modified proof modules remove 1,816 physical lines. The new coordinate helper costs 39, leaving 1,777 within the active subtree. The lower-level `PellRatio90.lean` dependency grows by six lines, giving the 1,771 net total. The generated Steps figure counts Lean source, not the Python generator changes. None is a compilation-speed measurement.

1.2 New construction expands the review surface

Complete old and current active Lean trees contain 608 and 637 files respectively: 29 additions, ten modifications, and no deletions. Additions comprise 14 construction modules, their 14 check modules, and the coordinate helper. New construction/check modules add 2,836 lines, and the umbrella adds 14 imports. Thus the complete active subtree grows by 1,073 lines despite the refactors.

The relevant fresh modules construct helical quotients and extensions, actual halted trace cycles, finite computation bands, marked canonical words, and soundness/completeness bridges. All five candidate target files are additions in this delta. Their newness helps locate untouched work but does not by itself establish novelty; the next sections impose the stronger semantic baseline.

Inventory caveat. The compare response contains exactly 300 repository file records and is capped. Active-source coverage instead comes from complete, untruncated Git trees. All 39 active changes appear in the capped list without discrepancy; no completeness claim is made for its other repository paths.

Replay evidence. `provenance/RECENT_COMMITS.md` provides the full commit list; `active_tree_delta.json`, `active_delta_crosscheck.json`, and `recent_adoption_metrics.json` in that directory record the authoritative accounting.

2 The baseline includes all promised prior outcomes

The user asked not to repeat proposals already in `research/lean-proof-simplification` or `docs/incoming`, and to assume they were applied. Maintained source alone is therefore an insufficient novelty test. A useful proposal must remove a residual obligation *after* prior promised outcomes and their advertised adapters are available.

2.1 What was inspected

At the pinned head, the research directory has 21 placed packets, `2026-10-26a` through `2026-10-26u`. The incoming directory has its historical README and no ZIPs. The audit rebinds 45 previously inspected manuscript/note blobs to the exact current Git identities and reads the nine new manuscripts l, m, o, p, q, r, s, t, and u with their substantive support. It also adds this review series' six round-three outcomes, whether or not separately placed upstream.

There are 1,140 inventoried placed blobs, 894 hash-bound placed files, and 279 document/technical-source bindings. This does not claim every historical archive or receipt was rehashed or inspected.

The resulting ledger has **168 overlapping rows in 25 semantic subsystems**: 115 inherited rows, 47 new or strengthened packet rows, and six prior-round rows. These are not 168 distinct simplifications. Repeated proposals, compatible alternatives, and strengthening relationships remain visible rather than summed as independent savings.

2.2 How the virtual baseline is used

1. Grant the advertised semantic interface, including production adapters explicitly promised by a prior packet. A prototype-only historical implementation does not make its already-proposed adapter fresh work.
2. Remove the proof obligations that interface was meant to remove. Search beyond the already simplified client, not for a new spelling of the same argument.
3. Preserve the original guards, data, witnesses, and axiom qualifications. An optional change that adds choice does not silently confer an unchanged axiom footprint.
4. When variants conflict as source edits, grant their compatible mathematical outcomes rather than pretending both bodies coexist. A research direction without a specified theorem reserves its theme; it does not create a fictitious universal API.

Metric distinction. The nine-line saving is measured against the authenticated *actual pinned files*. There is no materialized fully merged virtual source tree, so this is *not* a measured virtual-tree line delta. The incremental claim is semantic: the selected obligation is not already removed by the assumed prior outcomes. Its implementation cost is then reported honestly using complete actual-source candidate files.

2.3 Nearby ideas deliberately excluded

Residue/floor normalization, generic coordinate or word wrappers, finite tile dispatch and high-color fibers, first-failure/prefix selection, translated interval induction, and spatial shared-edge/live-configuration adapters all belong to the baseline. They are not renamed as new helical, trace, or soundness contributions here. Generic additive quotient architecture was previously considered and declined. The late packet additionally covers the concrete H1 and H2 clients, which are now excluded.

3 Packet coverage and incremental novelty

Every row below is granted, including its advertised adapters. Links resolve to the pinned research collection. "Reuse" means an earlier inspection rebound to exact current bytes; "new" means a newly reviewed manuscript. These are representative themes, not a replacement for the 168-row ledger.

Packet	Review	Prior outcomes already available
a	Reuse	PR congruence, list/assignment APIs, residuals, circuits, scanner/carry
b	Reuse	Witness blocks, validated histories, scanner and gate/circuit semantics
c	Reuse	Lookup/renaming, half-radix, coefficients, carrier tail, compiler/DAG
d	Reuse	Assignment naturality, coefficients, residues, finite-fiber pushforward
e	Reuse	Local algebra, lookup, internal-model bridge, finite dispatch
f	Reuse	Exact architectural interfaces, source windows, import organization
g	Reuse	Scanner simulation, carry normalization, folded masks, degree/row layers
h	Reuse	Coefficient bounds, exact decoder, subtraction/casts, witness transport
i	Reuse	Exact roots, decoder graph, inverse invariance, support and carry
j	Reuse	Decoder graph, signed congruence, internal views/fusion, graph depth
k	Reuse	Finite support, Pell parity, inverse pairing, Euclidean and resource APIs
l	New	Internal zero cases, Pell sign/order, integer norm, inverse invariance
m	New	Root descent, small Pell ordinate, signed relation, pairing/Boolean views
n	Reuse	Prior round 2 parent/decoder, word/prefix/cycle, NorthView, depth/CRT
o	New	South prefixes, aligned difference, high-color fibers, selector coherence
p	New	Pell ratio, variadic fragments, forks, support, exact pairing witnesses
q	New	Internal natural cases, norm obstruction, residue, minima, set closure
r	New	Pell base order, perturbation/error budget, half quotient, rule schedule
s	New	Object specs, scaled Pell reuse, integer recovery, certificate closure
t	New	List codec/parent transfer, local coefficient fibers, Pell, first exit
u	New	Pell allowance, rank bound, Int interval, first failure, spatial edge/trace

Also granted are the six prior-round results: sentinel-last coding, action-halt reflection, old-write-site color, absolute trace bounds, guarded integer intervals, and paired spatial edges. The incoming README identifies the same retired packets; no incoming ZIP existed at that initial pin.

One remaining candidate. B2 is paired forward construction from an actual halted configuration. H1/H2 were initially retained, but late packet C/D cover those exact mechanisms. B1 is also excluded: packet w's cyclic shared-edge grid removes temporal matching, including reset-to-initializer and the trace cycle's last row. A shorter local proof does not restore novelty once that advertised interface is granted. The table above records only the original a–u audit; later exclusions are separate.

Rejected neighbors. Local-cell source-window repackaging, first-halt/prefix selection, and a `runTile` shared-adapter bundle receive no new credit. The full decisions and conflict rules are in `baseline_audit/NOVELTY_DECISIONS.md` and `INTEGRATION_CONFLICTS.md`.

4 Late refresh removes three initial candidates

The initial source and corpus audit was pinned at `e652d21c413de59e4821f6e8e9b6a3a077bf5bfe`. During final review, additional packets arrived. A closing check at `09d65a93c9df8684c1a226341d6a5120623d1cdb`, recorded at 18:01:35 UTC, found 22 further commits, research packets a–w, and four incoming ZIPs. The late audit inspected those ZIPs and their exact Git-blob bindings. The retained B2 target source is unchanged at that later pin. This is a focused closing novelty check, not a claim to have repeated every original source analysis on the later tree.

ID	Final status	Reason
H1	Excluded	Late refinement C removes the same signed helix induction through periodicity. A product-action version instead of a scalar orbit is not a fresh mechanism.
H2	Excluded	Late refinement D supplies the same exact quotient certificate and the same existential and canonical consumers.
B1	Excluded	Packet w's cyclic shared-edge grid already removes temporal matching, including reset to initializer and the trace cycle's last row.
B2	Retained	One private halted-color witness proves both forward pass/finish allowedness facts. The late screen found no prior proposal removing this exact residual obligation.

Packet w's lines 480–507 supply the temporal architecture but explicitly leave literal allowedness. Its lines 577–583 also leave finite-table membership. This is the narrow distinction supporting B2, not permission to repeat the temporal architecture.

4.1 Exact overlap evidence

Four Further Lean Proof Refinements records closing pin `f42d3f94f78eb63639316a33e25f2f69d12c651e`. Its received ZIP has SHA-256

`bab188a31d586762212096523ab818c9135824b6d443b2a577827bf8759583c4`.

The exact overlap is visible in its `patches/helical-rigidity.patch` and `patches/exact-band-witness.patch`. All 82 listed artifact hashes agree with that archive's manifest. This verifies the received bytes, not its reported compiler execution. Its reported compilation and axiom results do not certify any differently written candidate in this report.

The local supplement `late_review/H1_H2_EXCLUSION.md` preserves this comparison and overrides the original positive H1/H2 decisions. The later `late_novelty_review/B1_B2_DECISIONS.md` records the B1 exclusion and B2 retention. Earlier audit counts, ledgers, and test receipts are retained as dated evidence, not silently rewritten into a claim that all later packets had been available initially.

4.2 How to use the comparison artifacts

The archived H1/H2/B1 code is excluded comparison evidence, omitted from the main manuscript. It explains exactly why the initial shortlist was reduced and permits replay of the original static checks. Do not apply those patches as additional simplifications after granting prior work. Only B2 contributes to the final nine-line estimate. The strict prior-applied requirement is more important than maintaining an arbitrary number of proposals.

5 B2 Share one halted color witness

Source anchors. `WangTiles942FinishConstruction.lean` lines 29–53; literal color and halted-head equations in `WangMachineEncoding.lean` lines 62–64 and `WangMachineEncoding.lean` lines 88–90; unchanged closing constructors in `WangTiles942Language.lean` lines 128–136.

The pass and finish membership proofs independently split head from nonhead and build the same tape-bit witness. Derive that witness once and directly produce both `Allowed` facts. Keep the helper private because no broader public color-normal-form API is needed for these two callers.

Complete replacement region.

```
/-- The halted color is derived once for the two literal closing constructors. -/
private theorem halted_tiles_allowed (cfg : Config 215) (hh : cfg.state = none) (x : ℤ) :
    Allowed (passTile cfg x) ∧ Allowed (finishTile cfg x) := by
  let b : Fin 2 := ⟨(Tape.read cfg.tape x).toNat, by
    have h := bit_le_one (Tape.read cfg.tape x); omega⟩
  simp only [passTile,finishTile,color,hh,halted_head_color]
  split_ifs
  · exact ⟨Allowed.pass (1 : Fin 2) b,Allowed.finish (1 : Fin 2) b⟩
  · simpa only [Fin.val_zero,Nat.mul_zero,zero_add] using
      And.intro (Allowed.pass (0 : Fin 2) b) (Allowed.finish (0 : Fin 2) b)

theorem passTile_allowed (cfg : Config 215) (hh : cfg.state = none) (x : ℤ) :
    Allowed (passTile cfg x) := (halted_tiles_allowed cfg hh x).1

theorem finishTile_allowed (cfg : Config 215) (hh : cfg.state = none) (x : ℤ) :
    Allowed (finishTile cfg x) := (halted_tiles_allowed cfg hh x).2
```

5.1 The exact finite color argument

The bit witness b lies in $\mathrm{Fin}\,2$ by `bit_le_one`. At a halted head, the constructor's head flag is $a = 1$ and the color is $432 + b$. Away from the head, $a = 0$ and the color is b. Thus the only input colors are $0, 1, 432, 433$. For the same (a, b), `Allowed.pass` keeps the color and `Allowed.finish` changes the south color to 438.

The conclusion is the conjunction of the two existing facts, not a new external premise or a changed tile language. The actual halt guard is indispensable: the initializer's operating-head color 4 is not included. Every actual configuration, arbitrary tape, and signed position remains allowed by the old quantification; no normalized tape representation is assumed.

Full cost. The replaced 25-line region becomes 16 lines. The new private declaration occupies ten lines including its comment; its separating blank and both two-line public wrappers are included. The complete file falls from 164 to 155 physical lines. Both public statements and every executable definition remain unchanged.

Novelty boundary. Prior high-color fibers characterize *existing* allowed tiles, and prior halt reflection identifies action state. Neither supplies this paired forward construction of the two actual halted closing rows. This result does not claim another finite-language reflection or general dispatch abstraction.

Tradeoff and gate. Lean must reduce the concrete `Fin 2` projections and local bit witness in the restricted simplifier. The four-color sanity model checks the constructor formula only; it is not a fresh exhaustive check of the 942-tile language. **Candidate only.** The Lean code in this report has not been compiled.

6 Evidence and the remaining Lean gate

6.1 What has actually been checked

The snapshot contains 914 selected current repository files, including all 637 active Lean files; six selected lower dependencies; and 11 prior-pin comparison files. Every selected file is bound by Git blob SHA-1 and SHA-256. These are selected-source and dependency snapshots, not a full checkout. The Mathlib periodicity excerpt is separately bound to its recorded dependency commit.

The original comparison replay applies all four patches together in one disposable source copy and reproduces all five complete candidate modules. The three supplied generators reproduce their 21 outputs byte-for-byte. Lexical declaration checks preserve old public signatures and complete executable definitions, with comments and whitespace normalized; the canonical word definition also has an exact byte-preservation check. These checks do not establish definitional equality in Lean for the new proof terms.

Evidence	Scope and limitation
H1 and H2 tests	Ten passing source/replay/API/model tests. Signed translation and independent-dimension quotient models do not prove actual tile membership or matching.
B1 chronology	525 symbolic seams for heights 3–32. Distinct actual-run times stay opaque; failing-height controls 1 and 2 remain. This is branch chronology, not machine execution.
B2 colors	Formula sanity check for $0, 1, 432, 433$; operating color 4 excluded. Not an enumeration or verification of the entire tile language.
Independent suite	60 passing tests include signed/zero-width cases, right-wall exclusion, height-0/1/2 controls, tampering, missing or empty axiom output, overlay order, and safe replay refusal. No Lean execution.
Independent review	No semantic gap found in the one retained and three now-excluded mechanisms. Original four-patch composition is a comparison check, not a recommendation to reapply prior work.

6.2 Required integration checks not performed here

Use an authorized complete checkout with the recorded toolchain. Apply only B2 to the assumed-updated baseline, compile its module and downstream checks, then compare baseline and patched `#print axioms` at the existing endpoints. Record commands, outputs, exit status and any repair overhead. No supplied script installs or fetches this environment.

For comparison reproducibility, the original route orders nine modules: five edited files (including excluded H1/H2/B1) and four unchanged connectors. It appends 96 public-theorem axiom commands. This is not the selected one-patch integration plan, nor 96 observed axiom results. All new candidates remain uncompiled; historical receipts and PDF rendering do not establish their elaboration, axioms, or performance.

7 Artifacts and focused next questions

Paths are relative to the archive root. TeX caches and page-preview images are not required deliverables.

Artifact	Purpose
`report/proof_simplification_report.tex` and `.pdf`	Editable manuscript and rendered report; companion preamble and section files are included.
`report/build.sh`	XeLaTeX build script with local-cache fallback; rebuilds the report only.
`helical/patches/` and `helical/proposed/`	H1/H2 excluded comparison diffs and modules, not selected changes.
`bridges/patches/` and `bridges/candidate/`	B2 selected diff/module; B1 excluded comparison.
`helical/README.md` and `bridges/README.md`	Mechanisms, source anchors, full-cost accounting, exclusions, and unexecuted Lean gates.
`baseline_audit/`	All-applied semantic ledger, source bindings, packet coverage, and candidate novelty decisions. The late supplement overrides H1/H2/B1 retention; historical bulk archives are not needed.
`provenance/` and `sources/`	Pinned manifests, commit/delta accounting, integrity checks, and selected authenticated sources.
`dependencies/`, `comparison_sources/`	Current lower dependencies and earlier files used for exact observed-adoption metrics.
`validation/`	Independent replay, ordered same-world Lean audit material, and static validation receipts. Audit commands remain unexecuted.

7.1 Reproduce the non Lean checks

Run from the extracted archive root, using Python 3 and the recorded scripts:

```
python provenance/verify_snapshot.py
python validation/source_validate.py
python helical/validation/test_candidates.py
python bridges/validate_static.py
bash report/build.sh
```

The Python commands check evidence or candidates without Lean. The last command needs XeLaTeX and rebuilds only the PDF. Read each component README before rerunning candidate generators; their inputs are the pinned snapshot, not a moving branch.

7.2 Questions worth resolving after elaboration

These are research questions, not additional selected ideas or savings.

1. Does the retained proof elaborate unchanged after all compatible prior APIs are integrated? Preserve the halting guard, and charge any repair or helper overhead.
2. Does the shared halted-color helper remain worthwhile if future pass and finish constructors diverge? Its private status permits a local adjustment without changing the public API.

7.3 Additional directions screened but not selected

A closing screen examined `CurrentWordTopBounds2011`, `CurrentWordCopies2011`, `CurrentWordLowMask2011`, and `ControllerResidueLift`. Prefix/support bounds, copy/parity adapters, and origin-erasure or no-borrow interfaces are already granted by prior proposals. The residue lift already uses a direct quotient/remainder argument; local cast cleanup did not justify another recommendation. Exact source hashes and reasons are recorded in `late_novelty_review/BOUNDED_RESIDUAL_SCREEN.json`. This bounded search found no robust second candidate; it is not a proof that no further simplification exists.